VINAY KUMAR

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EDUCATION

New York University Aug 2025 – May 2027
Master of Science in Computer Science *GPA: 3.67/4.0*
- Coursework: Deep Learning Systems, Machine Learning for Signal Processing, Big Data Analytics, Distributed Systems, Advanced Algorithms

BNM Institute of Technology 2021 – Apr 2025
Bachelor of Engineering in Artificial Intelligence and Machine Learning *CGPA: 8.69/10*
- Coursework: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Data Structures, Operating Systems

EXPERIENCE

NYU Information Technology (NYU IT) Jan 2026 – Present
Learning Management System (LMS) Specialist *New York, NY*
- Architected autonomous agentic workflows using Python and LLM APIs to ingest, summarize, and categorize unstructured faculty consultations
- Built automated prompt-chaining evaluation scripts with strict JSON schema outputs to triage technical incidents, cutting manual routing overhead

Healium Intelliscan Mar – May 2025
Machine Learning Systems Intern *India*
- Constructed computer vision pipelines in PyTorch for renal ultrasound analysis, applying CLAHE across 80K frames and augmentations across 40K scans
- Trained deep feature extraction networks (EfficientNet-B7) and gradient-boosted ensembles (XGBoost), boosting out-of-distribution F1 score by 14%
- Engineered distributed data loaders and batch inference routines using ONNX Runtime, reducing end-to-end model latency by 79%

KreditBee Sep – Dec 2024
Machine Learning Intern *India*
- Trained and deployed real-time identity anti-spoofing classifiers in Python and TensorFlow/Keras across high-concurrency verification streams
- Formulated hard-negative mining routines and multi-task loss functions across face embeddings, reducing presentation spoof attack error rates by 76%
- Optimized inference throughput by quantizing convolutional model weights to INT8, achieving sub-45ms execution on edge CPU nodes

369 Data Solutions Sep – Dec 2023
Machine Learning Intern *India*
- Trained and evaluated ResNet and VGG backbone models in PyTorch for automated defect verification across high-variance crystallization imagery
- Engineered confidence-calibration and out-of-distribution rejection layers that curbed false-positive classifications by 85% across test batches

PROJECTS

Autonomous System Diagnostics & Telemetry Remediation Agent | *Python, LangGraph, FAISS, Docker* *2026*
- Engineered a multi-step agentic triage engine using LangGraph state graphs and Plan-and-Execute loops to trace distributed system failure cascades
- Constructed a semantic RAG pipeline indexing 115+ incident post-mortems in FAISS with dense cross-encoder reranking to ground hypothesis generation
- Integrated tool-calling interfaces (log parsers, metric telemetry analyzers) to run autonomous root-cause verification without model hallucinations

PromptDoctor: LLM Evaluation & Regression Analysis Workbench | *Python, React, Redis, Sidekiq, MCP* *2026*
- Built an end-to-end evaluation platform to benchmark prompt drift, latency trade-offs, and token economics across multi-provider LLM pipelines
- Implemented Anthropic Model Context Protocol (MCP) server endpoints to interface external testing harnesses and ground-truth validation datasets
- Architected asynchronous Redis worker queues to execute parallelized automated evaluation suites and capture structured diff analytics

TECHNICAL SKILLS

Core AI & ML: PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, XGBoost, ONNX Runtime, Hugging Face, NumPy, Pandas
LLMs & Agentic Systems: LangGraph, LangChain, FAISS, Model Context Protocol (MCP), Vector Databases, RAG Architectures
Languages & Backend: Python, C++, SQL (PostgreSQL, SQLite), Go, Java, Bash, FastAPI, RESTful APIs
Infrastructure & MLOps: Docker, Kubernetes, AWS, Redis, Git, Linux/Unix, CI/CD, Model Quantization, TensorRT